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S2**Started on** Thursday, 24 September 2026, 9:11 AM**State** Finished**Completed on** Thursday, 24 September 2026, 9:19 AM**Time taken** 8 mins 6 secs**Marks** 1.00/1.00**Grade** 10.00 out of 10.00 (100%)

Question 1 | Correct Mark 1.00 out of 1.00

Problem statement:

Find the length of the Longest Non-decreasing Subsequence in a given Sequence.

Eg:

Input:9

Sequence:[-1,3,4,5,2,2,2,3]

the subsequence is [-1,2,2,2,2,3]

Output:6

Answer: (penalty regime: 0 %)

```

1  #include<stdio.h>
2
3  int max(int a, int b) {
4      return (a > b) ? a : b;
5  }
6
7  int main() {
8      int n;
9
10     if (scanf("%d", &n) != 1 || n <= 0) {
11         return 0;
12     }
13     int arr[n];
14     int dp[n];
15     int maxLength = 0;
16
17     for (int i = 0; i < n; i++) {
18         if (scanf("%d", &arr[i]) != 1) {
19             return 1;
20         }
21         dp[i] = 1;
22     }
23     for (int i = 0; i < n; i++) {
24         for (int j = 0; j < i; j++) {
25
26             if (arr[i] >= arr[j]) {
27                 dp[i] = max(dp[i], dp[j] + 1);
28             }
29         }
30
31         if (dp[i] > maxLength) {
32             maxLength = dp[i];
33         }
34     }
35
36     printf("%d\n", maxLength);
37
38     return 0;
39 }
```

	Input	Expected	Got	
✓	9 -1 3 4 5 2 2 2 2 3	6	6	✓
✓	7 1 2 2 4 5 7 6	6	6	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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